

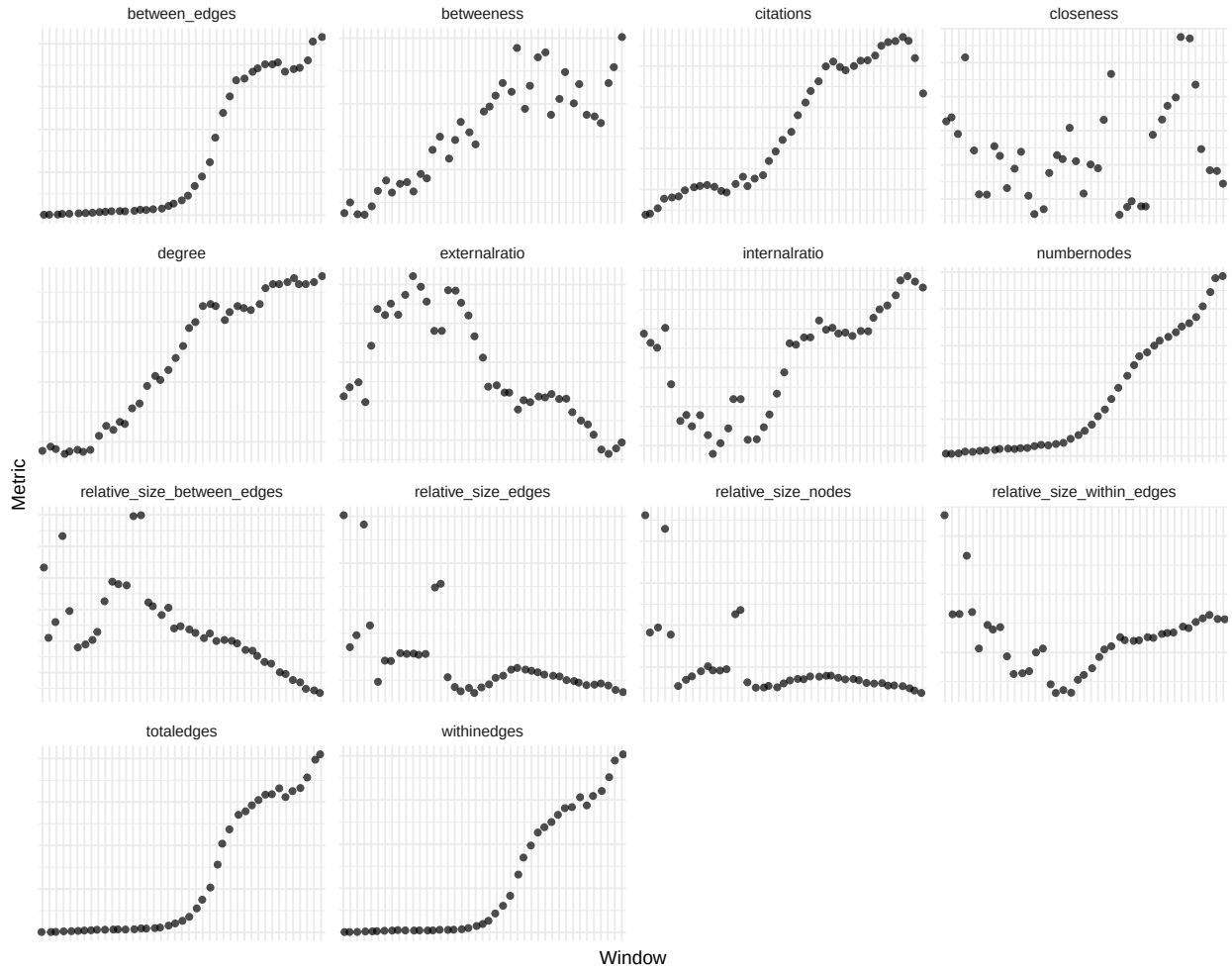
Field-Level Analysis

2025-11-20

1. Aggregate Metrics

The Metrics used here are formally defined on a per-field basis below. The aggregate metrics are computed similarly, but over all fields instead of per field.

1.1 Evolution Aggregate of Metrics



1.2 Number of Nodes and Edges over Time

Window	Number of Nodes	Number of Edges
1980-1984	71	203
1981-1985	85	301
1982-1986	104	377
1983-1987	135	582
1984-1988	163	842
1985-1989	233	1182
1986-1990	248	1385
1987-1991	270	1579
1988-1992	307	1900
1989-1993	316	1944
1990-1994	302	2083
1991-1995	339	2183
1992-1996	346	2085
1993-1997	364	1940
1994-1998	406	2433
1995-1999	477	2940
1996-2000	545	3390
1997-2001	598	3879
1998-2002	776	5356
1999-2003	948	7176
2000-2004	1128	9121
2001-2005	1388	12127
2002-2006	1754	18350
2003-2007	2052	24776
2004-2008	2493	33425
2005-2009	2985	50247
2006-2010	3501	66059
2007-2011	3962	76943
2008-2012	4380	88183
2009-2013	4575	91423
2010-2014	4863	96227
2011-2015	5102	100431
2012-2016	5316	105586
2013-2017	5531	106235
2014-2018	5778	111244
2015-2019	5974	105257
2016-2020	6244	109745
2017-2021	6746	111773
2018-2022	7436	120666
2019-2023	8124	136208
2020-2024	8282	141309

2. Evolution of Fields

Field-Level Metrics are derived from networks where articles are nodes and edges are based on the strength of bibliographic coupling between articles. The nodes, edges and their information is then aggregated per discipline, here called field.

2. Relative Number of Nodes and Number of Citations

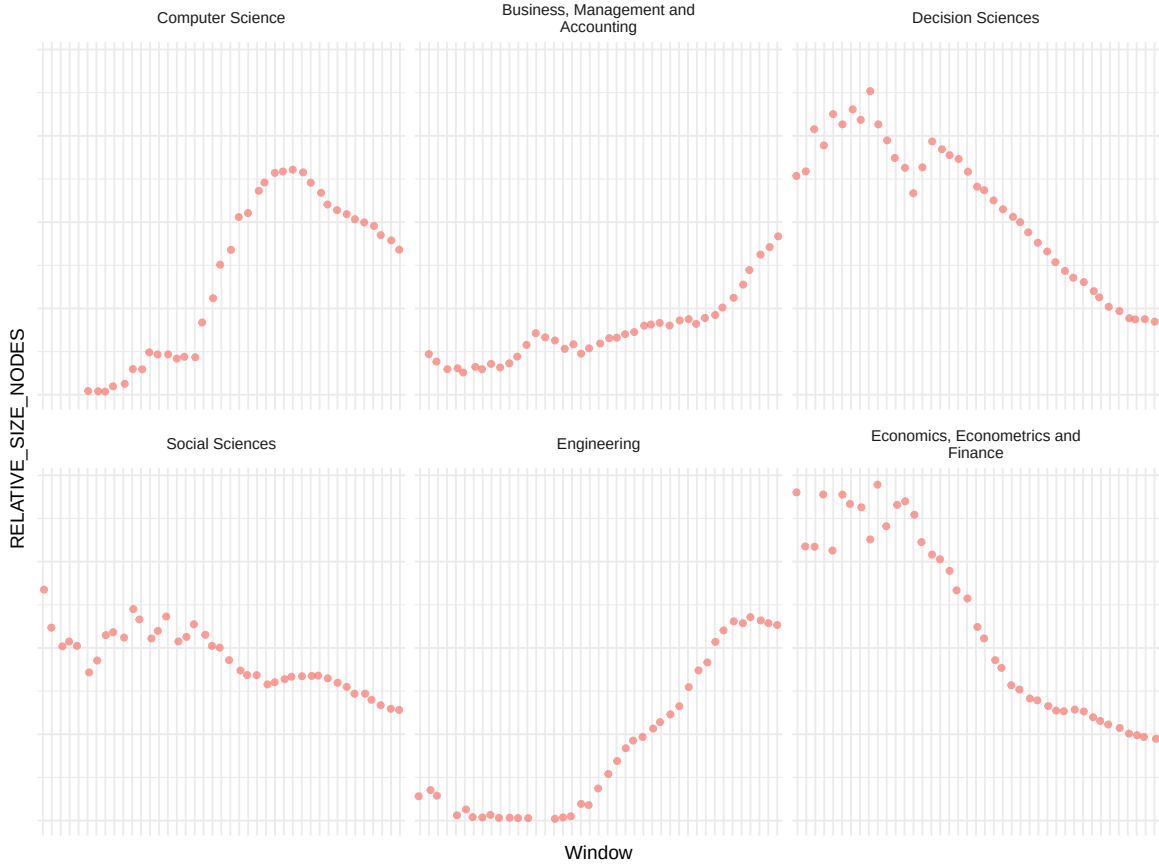
- Relative number of Nodes:

$$RelNodes_{t,f} = \frac{N(V)_{t,f}}{\sum_f^F N(V)_{t,f}} \quad (1)$$

where:

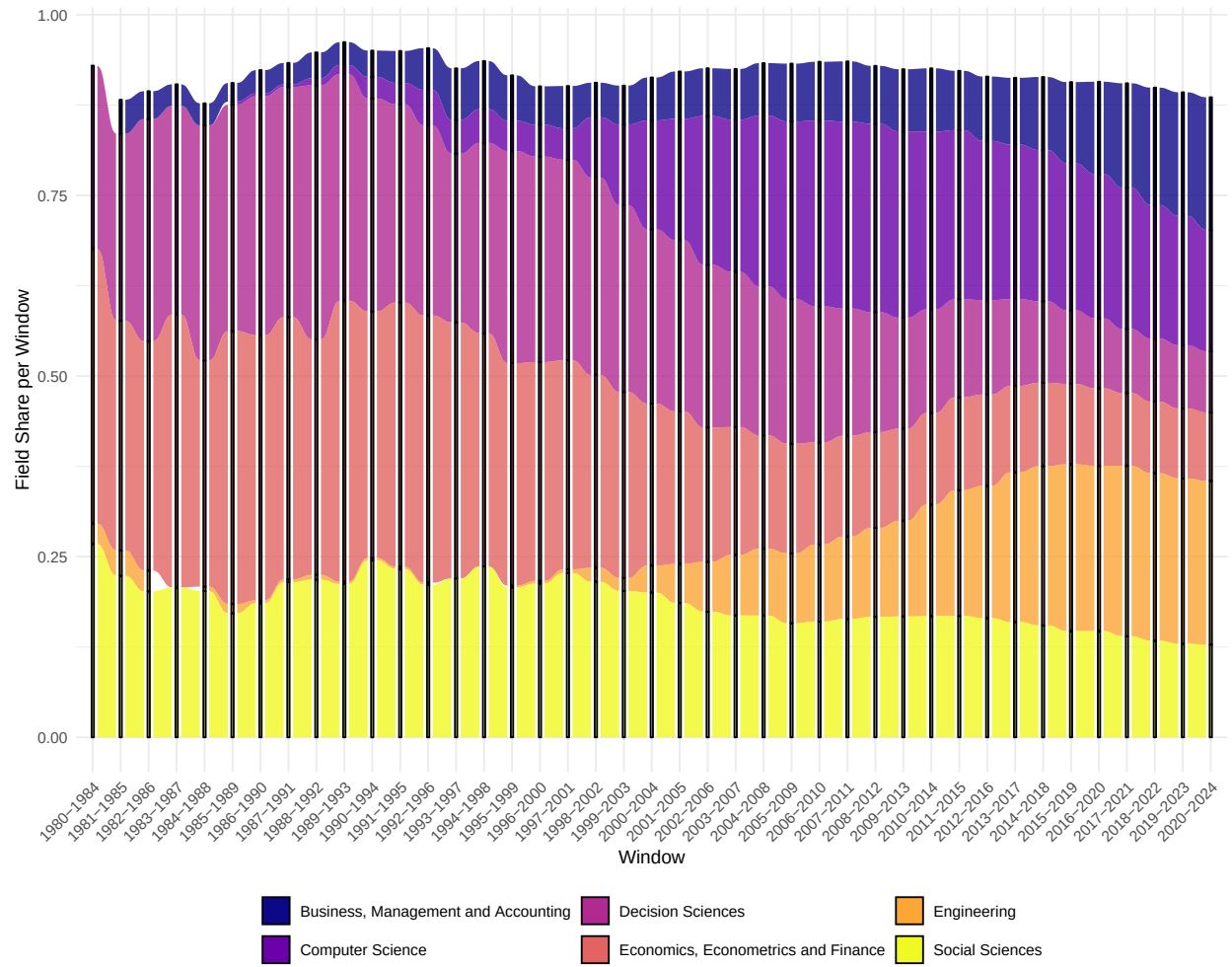
- $RelNodes_{t,f}$ measures the relative number of nodes for field f in time window t across all article nodes
- $N(V)_{t,f}$ measures the total number of nodes for field f in time window t across all article nodes. It is therefore defined as $N(V)_{t,f} = \sum_{v_i} 1_{field(v_i)=f}$, where $1_{field(v_i)=f}$ is an indicator function, taking the value one if article node v_i belongs to field f .
- The number of citations per field and time window is measured as the sum of citations across all articles of a field in a give time window, respectively.

Evolution of Relative Number of Nodes per Field



Evolution of Citations per Field





2.2 Relative Number of Edges

- Relative Number of Total Edges:

$$RelEdges_{t,f} = \frac{N(E)_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (2)$$

where:

- $RelEdges_{t,f}$ measures the relative number of edges incident on nodes within field f in time window t
- $N(E)_{t,f}$ measures the total number of edges incident on nodes within field f in time window t . It is therefore defined as $N(E)_{t,f} = \sum_{e=(v_i,v_j)} 1_{field(v_i)=f \vee field(v_j)=f}$, where $1_{field(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to field f
- Relative Number of Within-Field Edges:

$$RelEdgesWithin_{t,f} = \frac{N(E_{within})_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (3)$$

where:

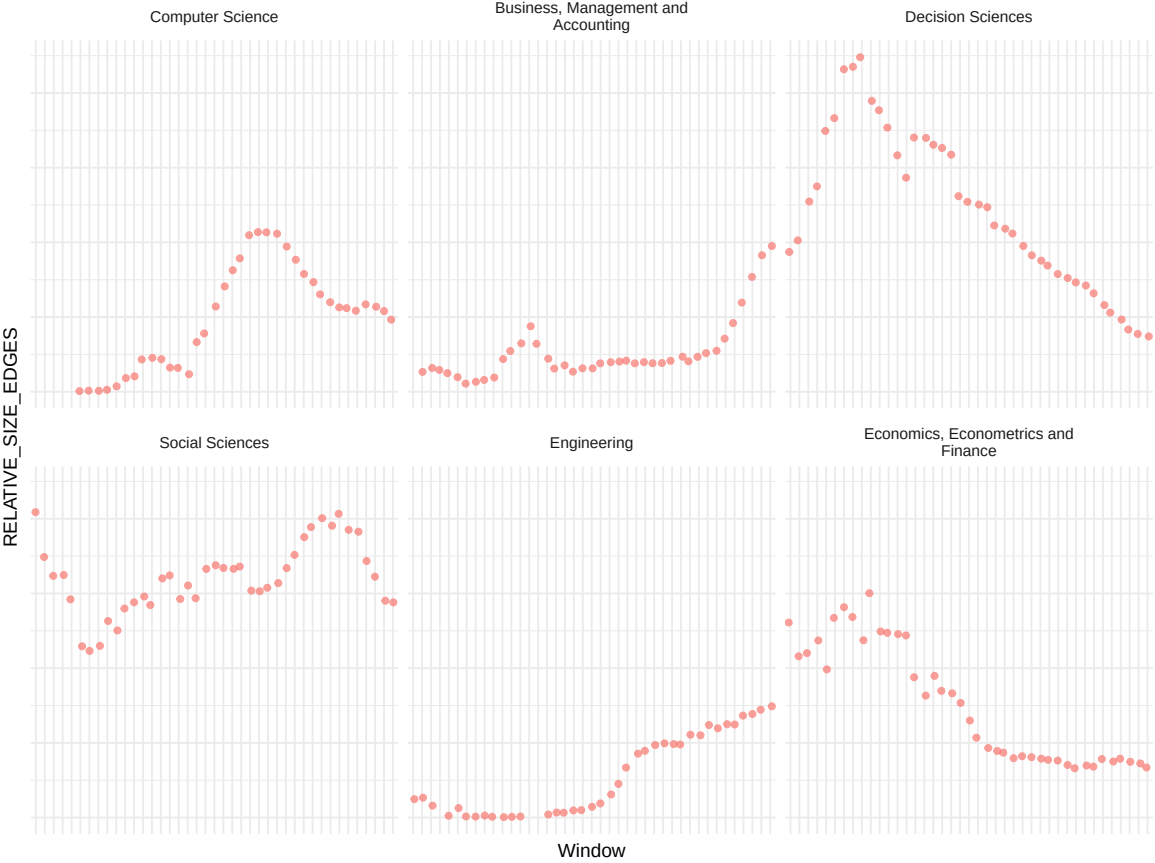
- $RelEdgesWithin_{t,f}$ measures the relative number of edges between article nodes of the same field for field f in time window t
- $N(E_{within})_{t,f}$ measures the total number of edges between article nodes of the same field for field f in time window t . It is therefore defined as $N(E_{within})_{t,f} = \sum_{e=(v_i,v_j)} 1_{field\{(v_i)=f\}} 1_{\{field(v_j)=f\}}$, where $1_{field(v_k)=f}$ is an indicator function, taking the value one if article node k belongs to field f .
- Relative Number of Between-Field Edges:

$$RelEdgesBetween_{t,f} = \frac{N(E_{Between})_{t,f}}{\sum_f^F N(E)_{t,f}} \quad (4)$$

where:

- $RelEdgesBetween_{t,f}$ measures the relative number of edges between article nodes of different fields for field f in time window t
- $N(E_{Between})_{t,f}$ measures the total number of edges between article nodes of different fields for field f in time window t . It's defined as $N(E_{Between})_{t,f} = \sum_{e=(v_i,v_j)} (1_{field\{(v_i)=f\}} 1_{field\{(v_j) \neq f\}} + 1_{field\{(v_i) \neq f\}} 1_{field\{(v_j)=f\}})$.

Evolution of Relative Number of Edges per Field



Evolution of Relative Number of Edges within/between Fields



2.3 Internal and External Edge Ratio

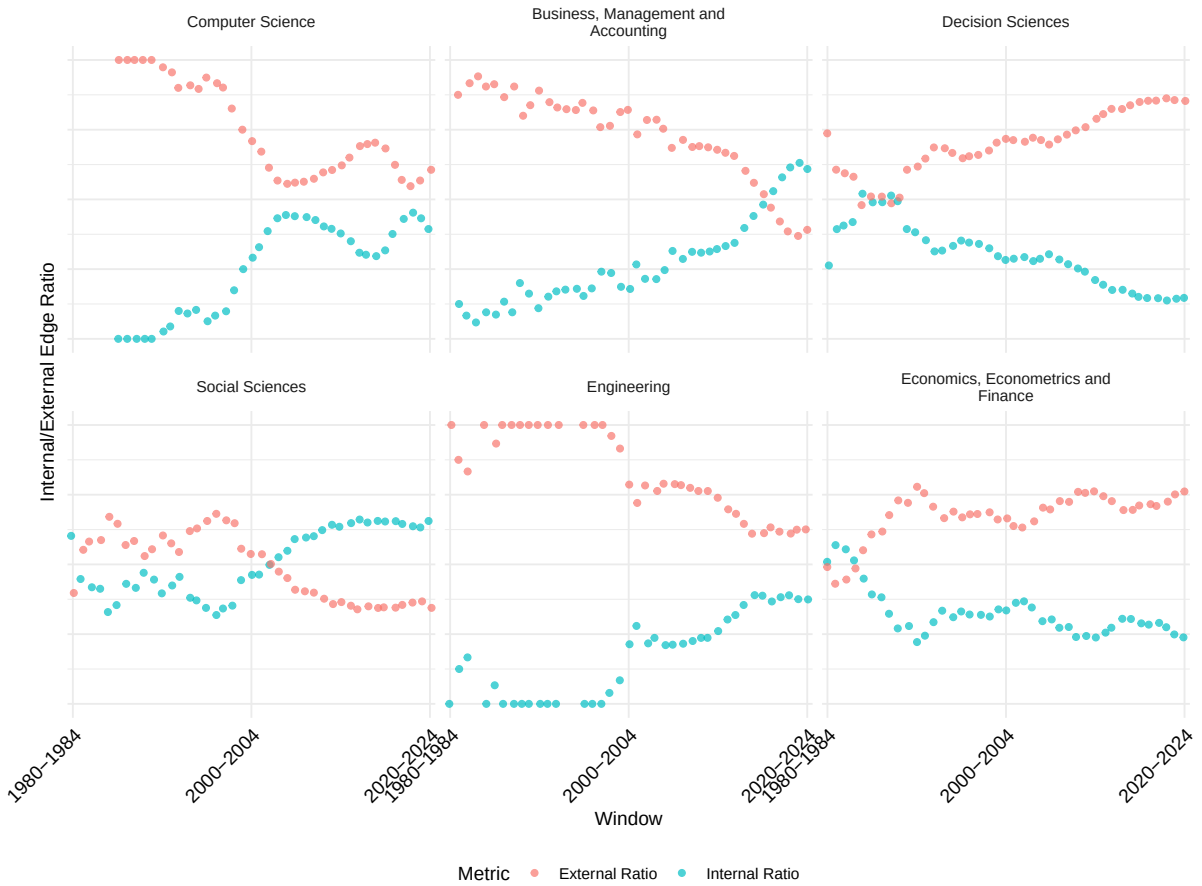
- Internal Edge Ratio:

$$IR_{t,f} = \frac{N(E_{within})_{t,f}}{N(E)_{t,f}} \quad (5)$$

- External Edge Ratio:

$$ER_{t,f} = \frac{N(E_{between})_{t,f}}{N(E)_{t,f}} \quad (6)$$

with $N(E_{within})_{t,f}$, $N(E_{between})_{t,f}$ and $N(E)_{t,f}$ defined as before. The internal and external ratios therefore measure the ratio of edges within (between) fields relative to the total number of edges of a field. Edges are derived from article-level nodes.



2.4 Network Centrality Metrics (Degree, Closeness and Betweenness)



Evolution of Field-level Closeness

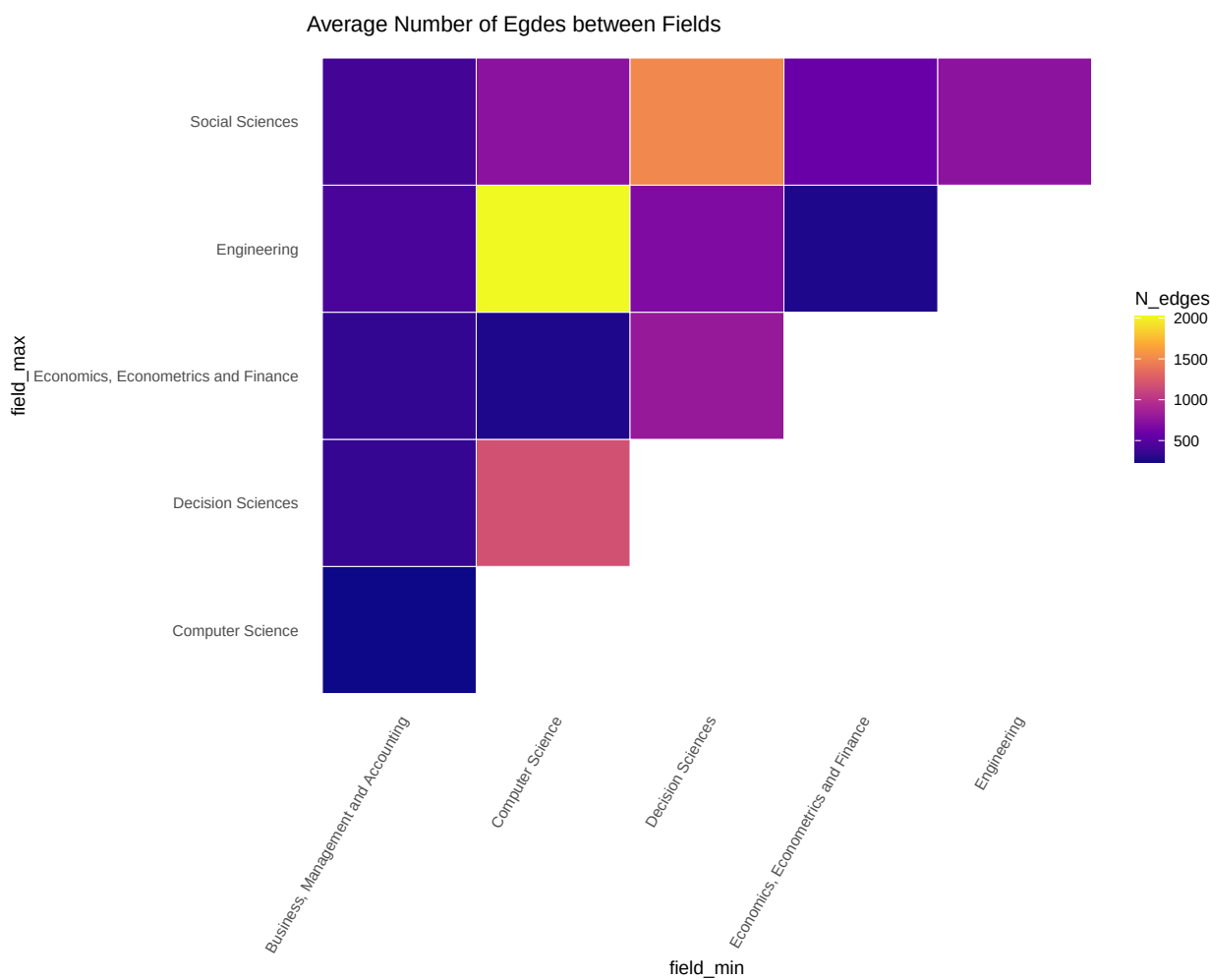


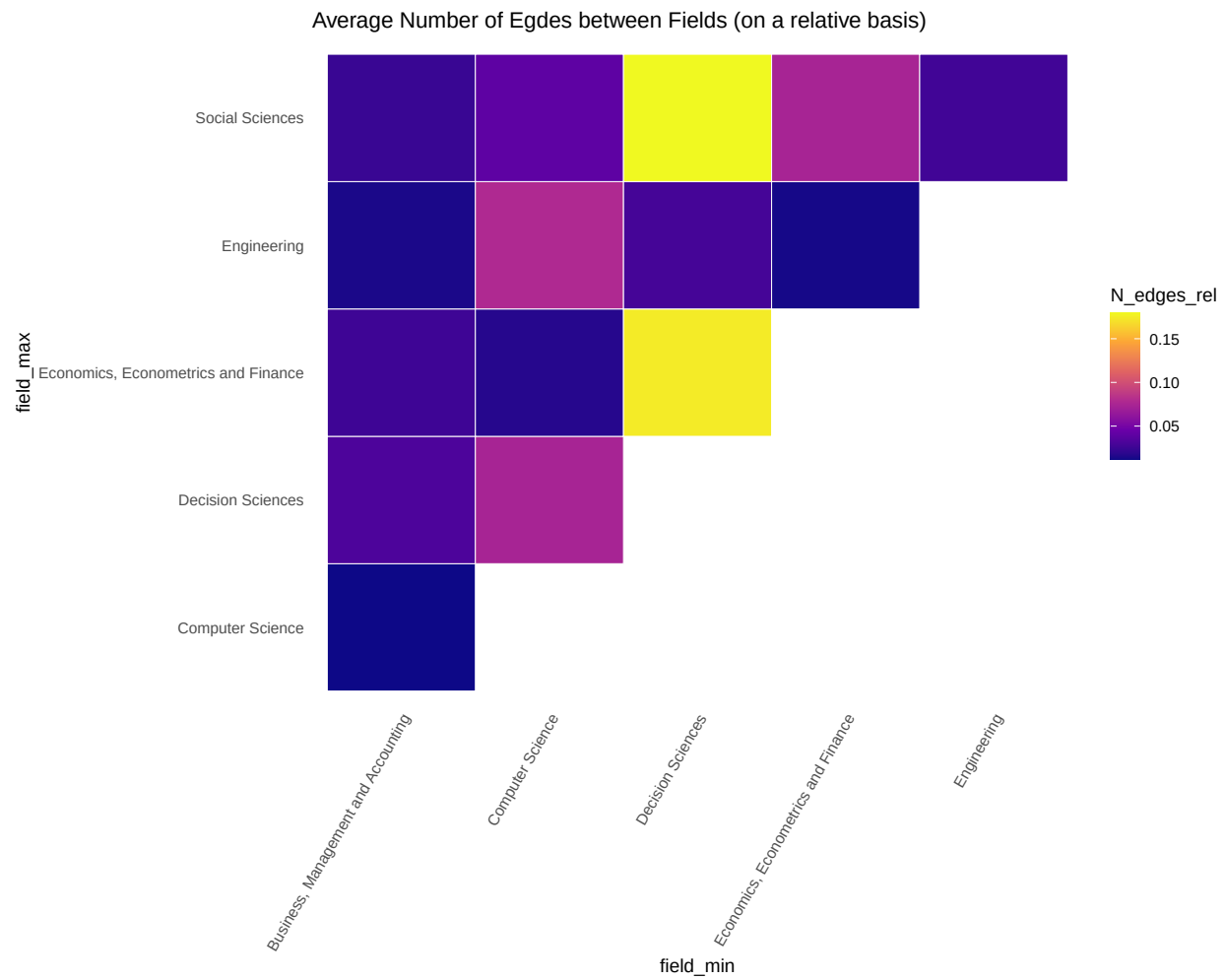
Evolution of Field-level Betweenness



3. Field-Field Connectivity

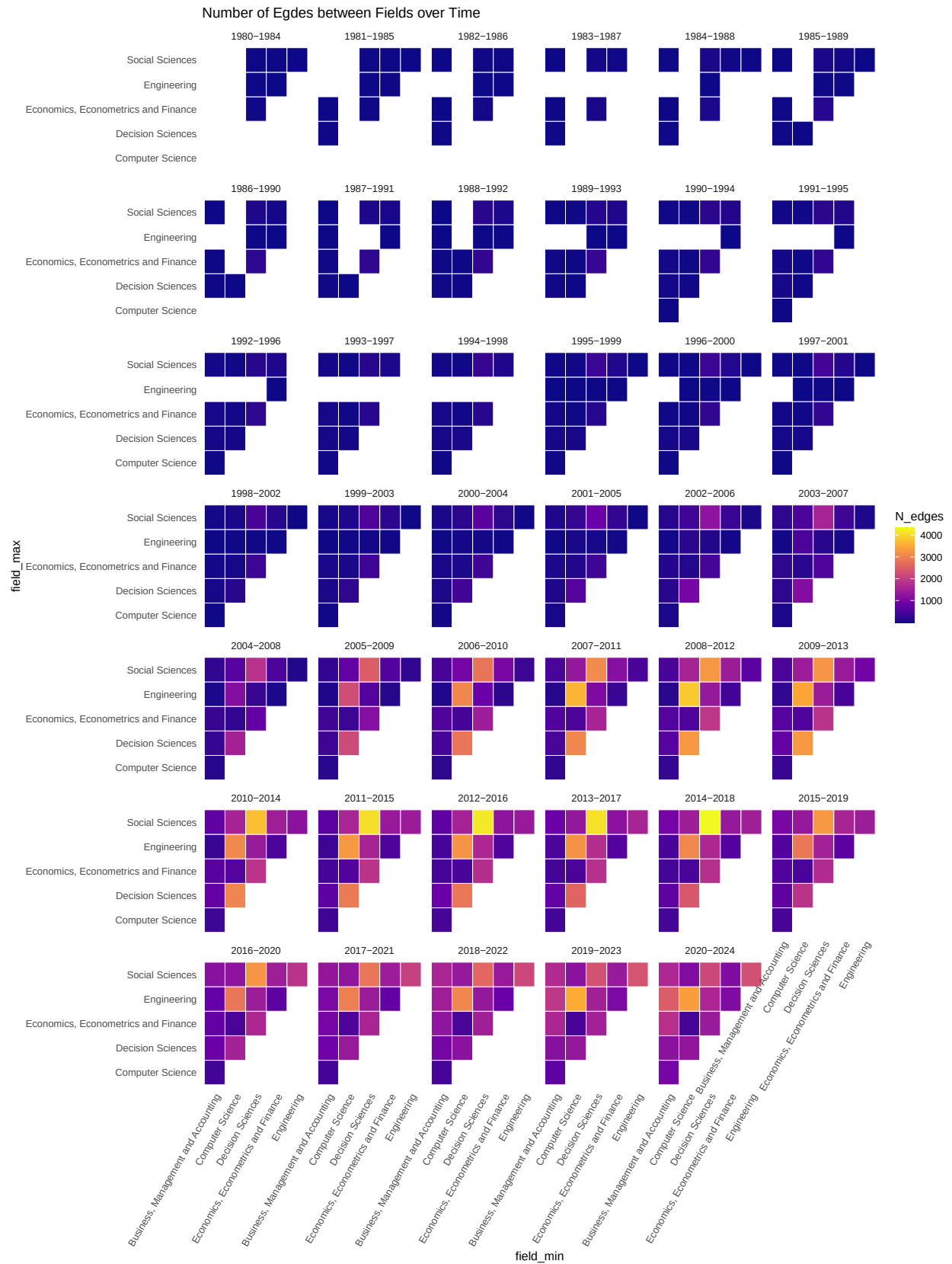
3.1 Average Connectivity

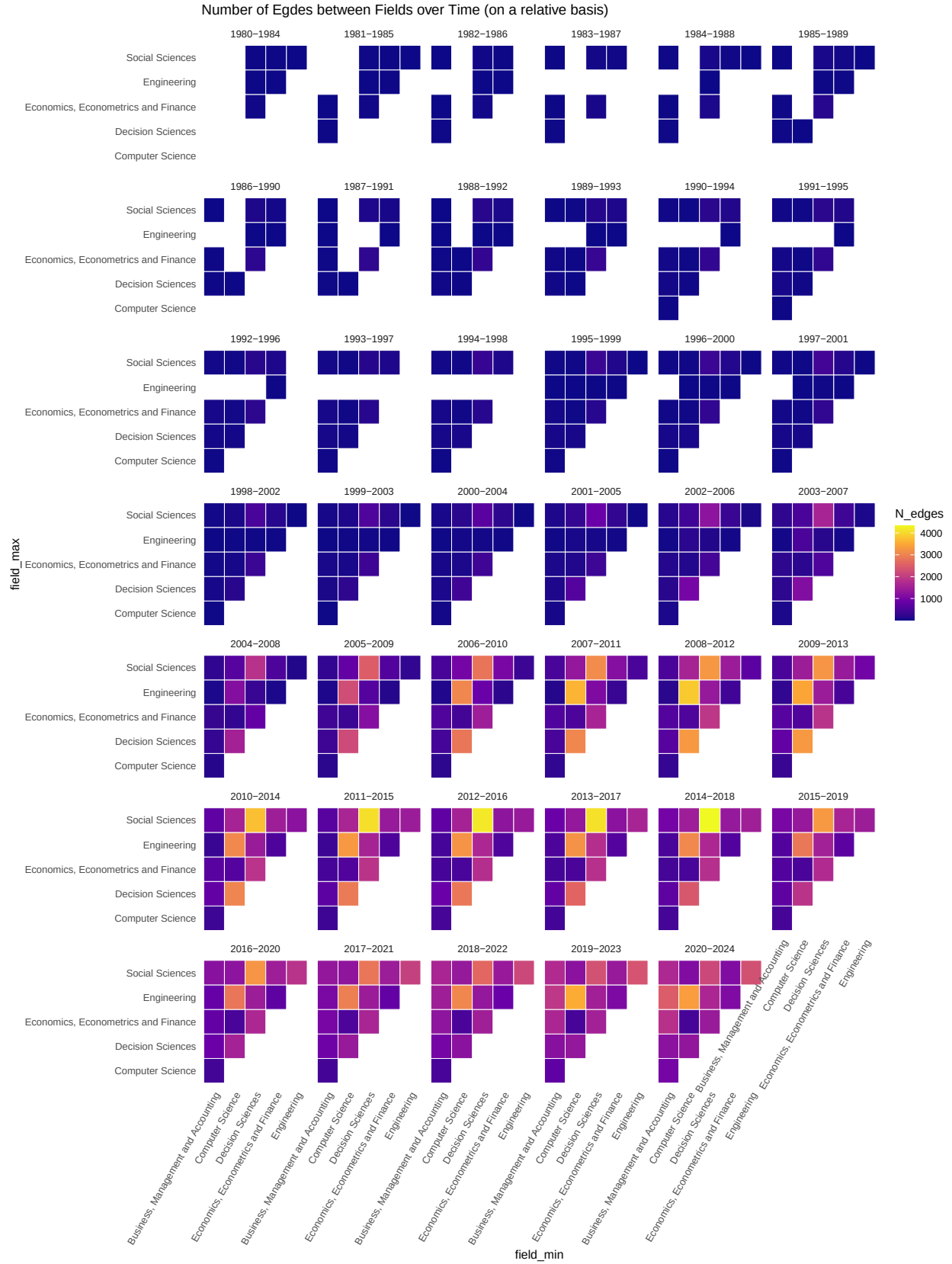




3.2 Connectivity over time

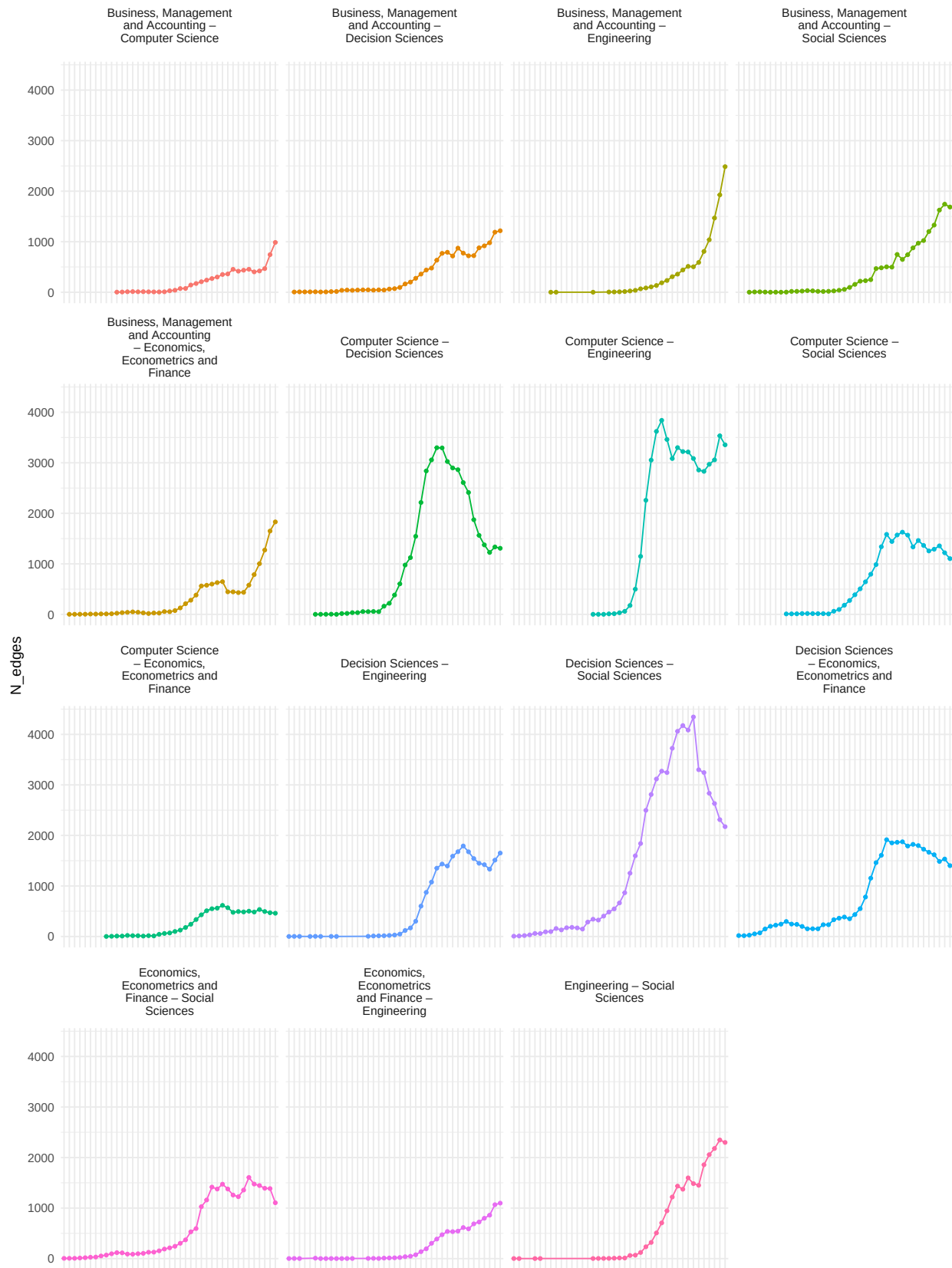
3.2.1 Heatmaps



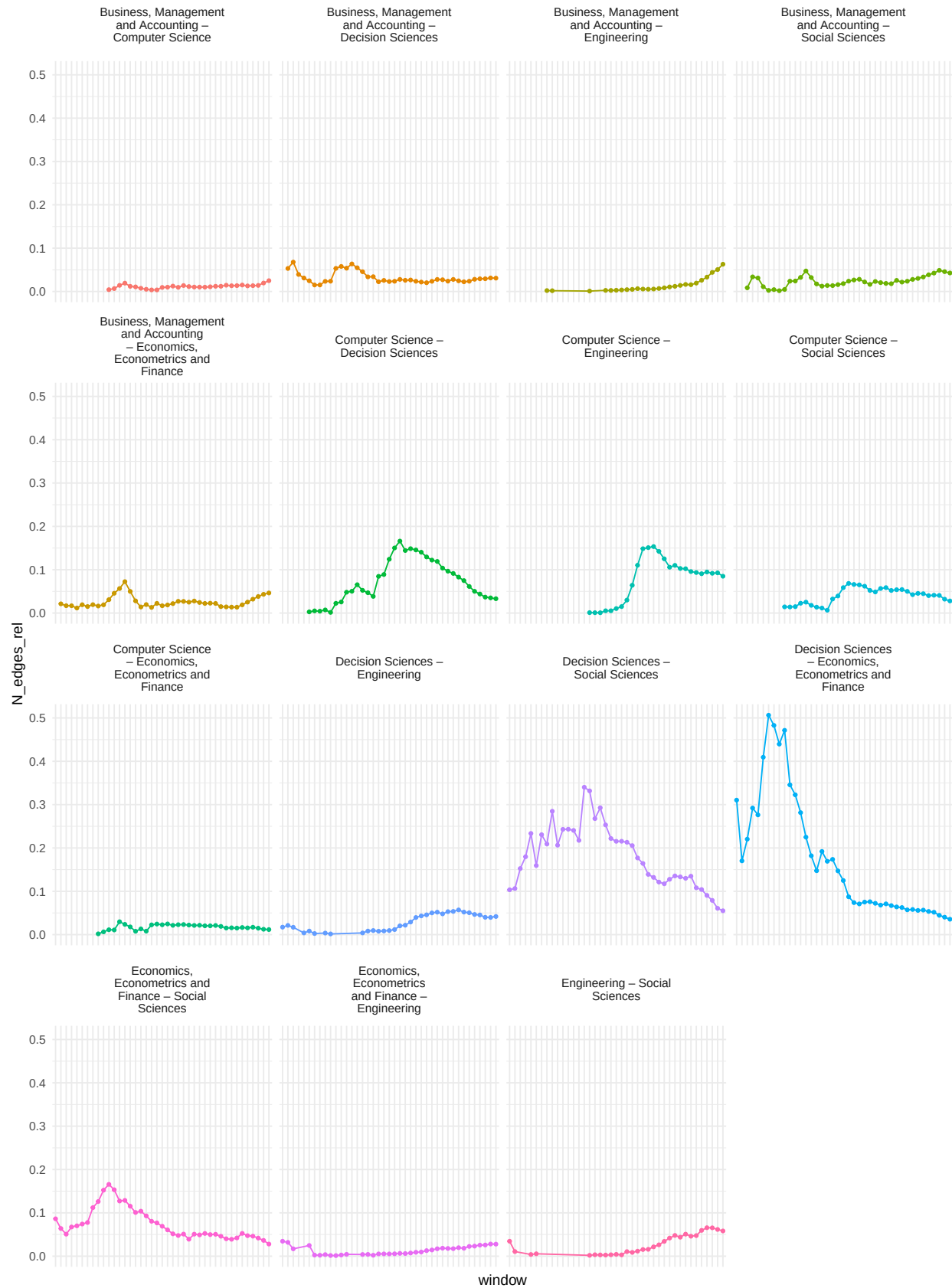


3.2.2 Time Series

Number of Edges between Fields over Time



Number of Edges between Fields over Time (on a relative basis)



4. Citation Flows

4.1 Internal Coverage

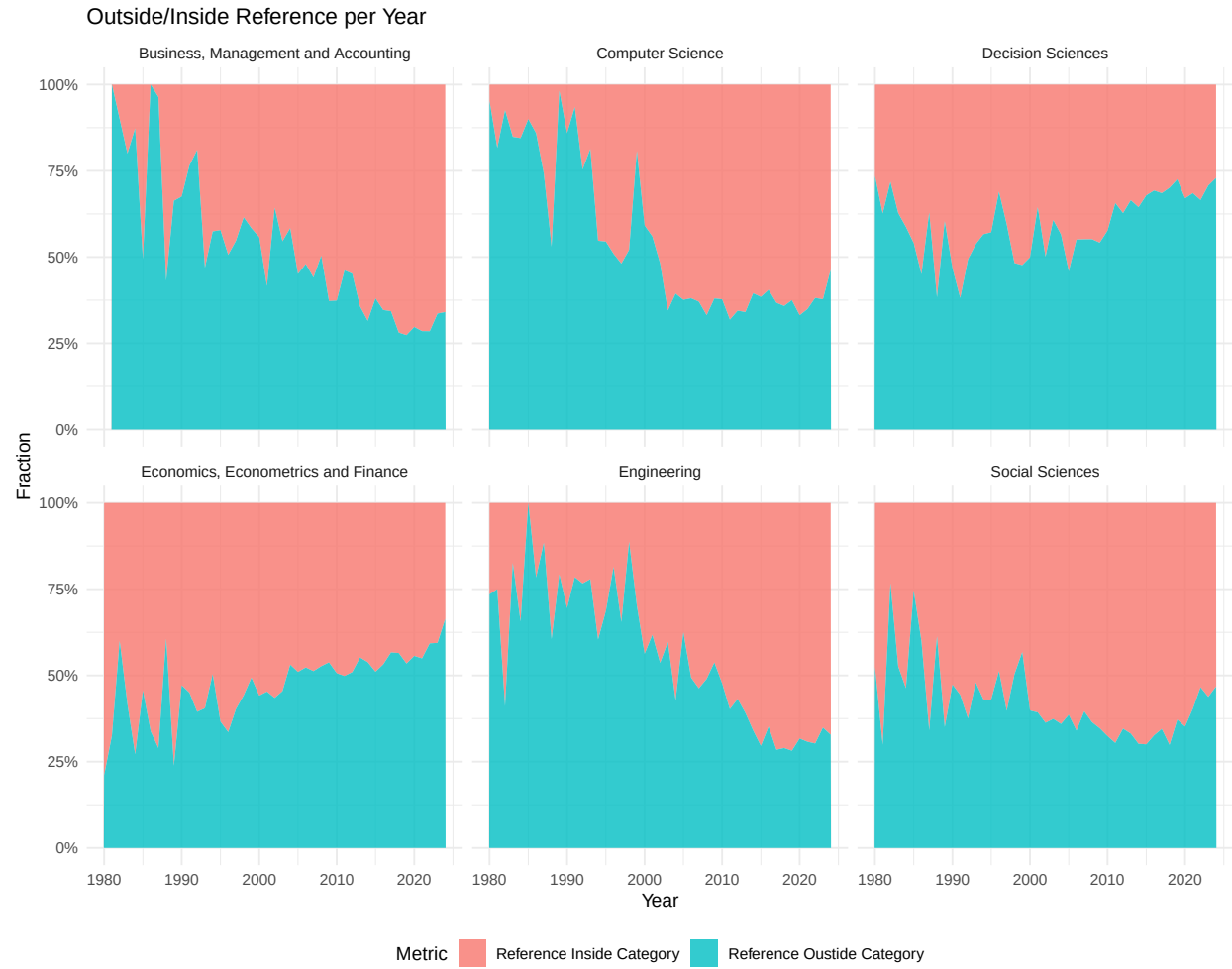
Table 1: Number of References with Meta Data

Field	Total Number of References	Relative Number of References (in %)
Agricultural and Biological Sciences	15878	1.45
Arts and Humanities	6257	0.57
Biochemistry, Genetics and Molecular Biology	15103	1.38
Business, Management and Accounting	122986	11.22
Chemical Engineering	101	0.01
Chemistry	587	0.05
Computer Science	172746	15.76
Decision Sciences	107728	9.83
Dentistry	24	0.00
Earth and Planetary Sciences	2806	0.26
Economics, Econometrics and Finance	134496	12.27
Energy	5772	0.53
Engineering	153172	13.98
Environmental Science	29274	2.67
Health Professions	4473	0.41
Immunology and Microbiology	867	0.08
Materials Science	777	0.07
Mathematics	17842	1.63
Medicine	10669	0.97
Neuroscience	5129	0.47
Nursing	172	0.02
Pharmacology, Toxicology and Pharmaceutics	158	0.01
Physics and Astronomy	20785	1.90
Psychology	14310	1.31
Social Sciences	146557	13.37
Veterinary	142	0.01
NA	106959	9.76

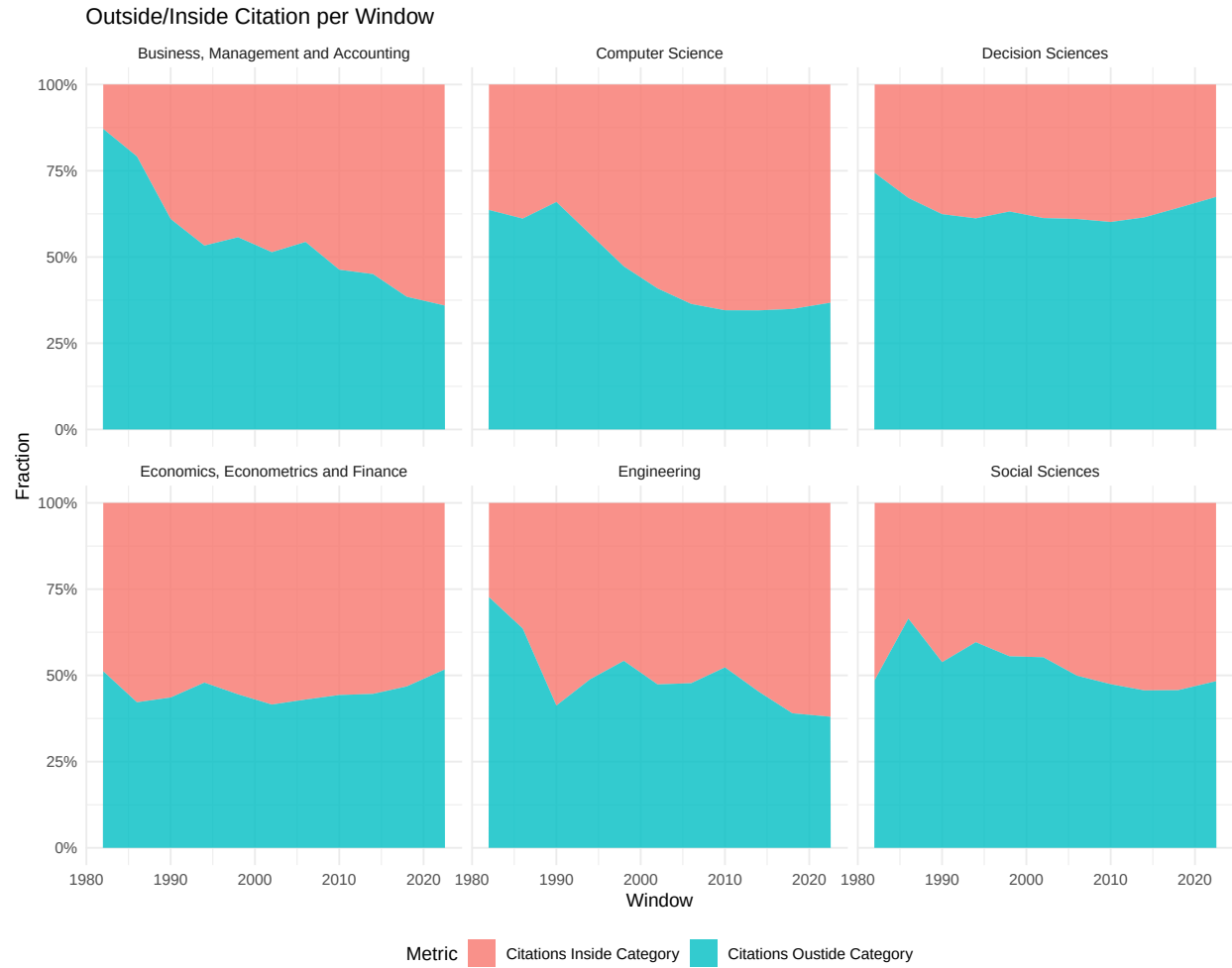
4.2 Outside/Inside Citations

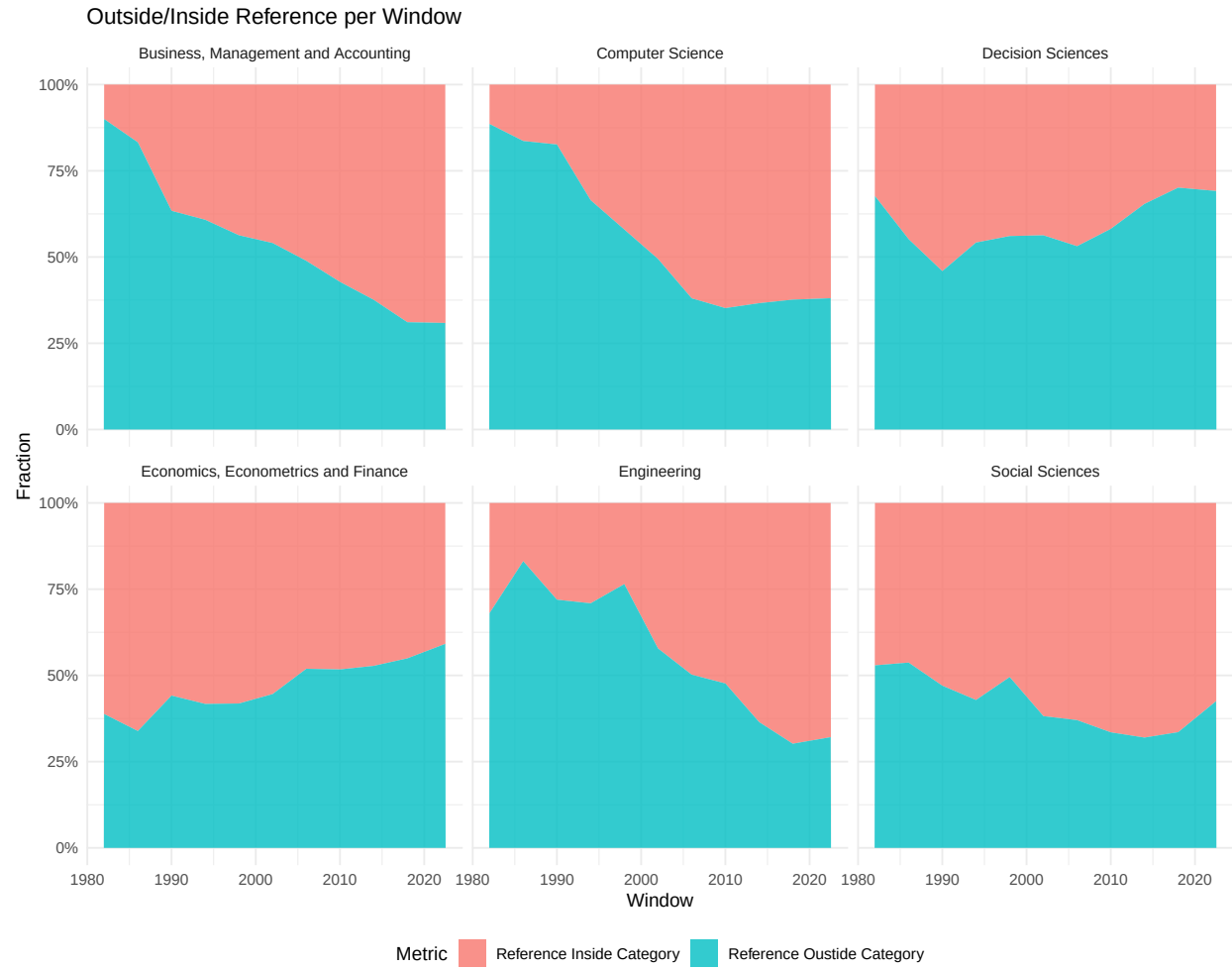
4.2.1 Per Year



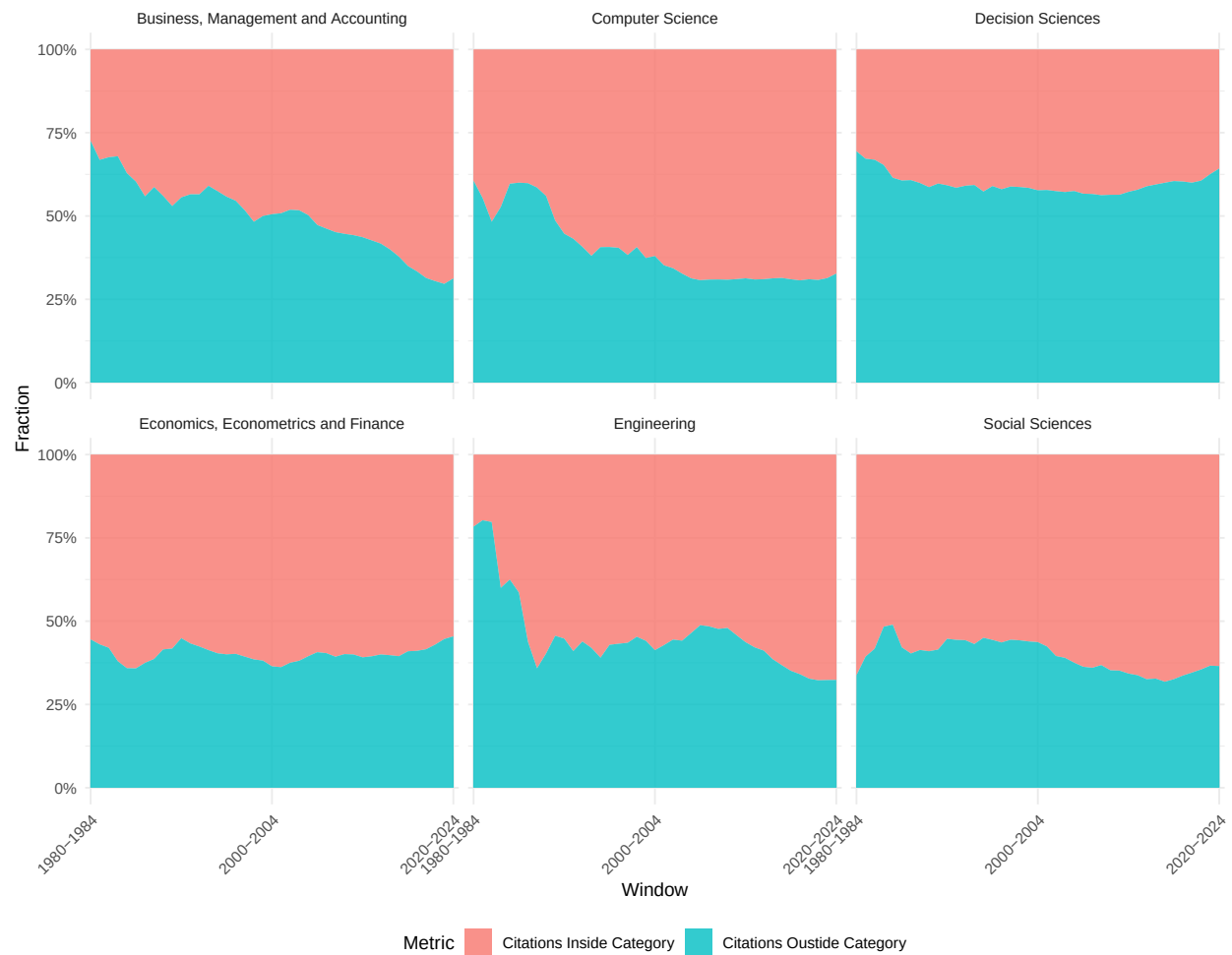


4.2.2 Per Window (non-overlapping)

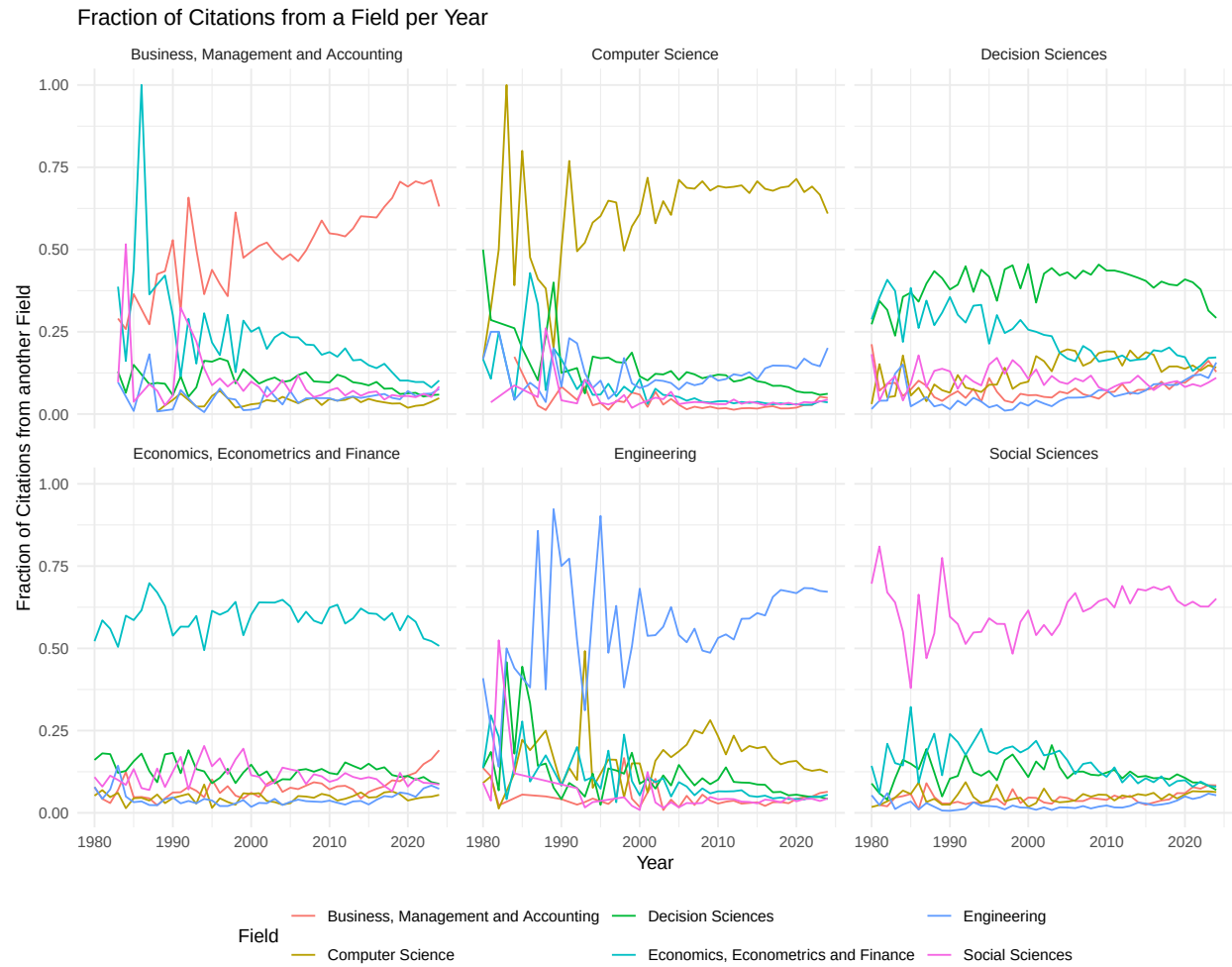




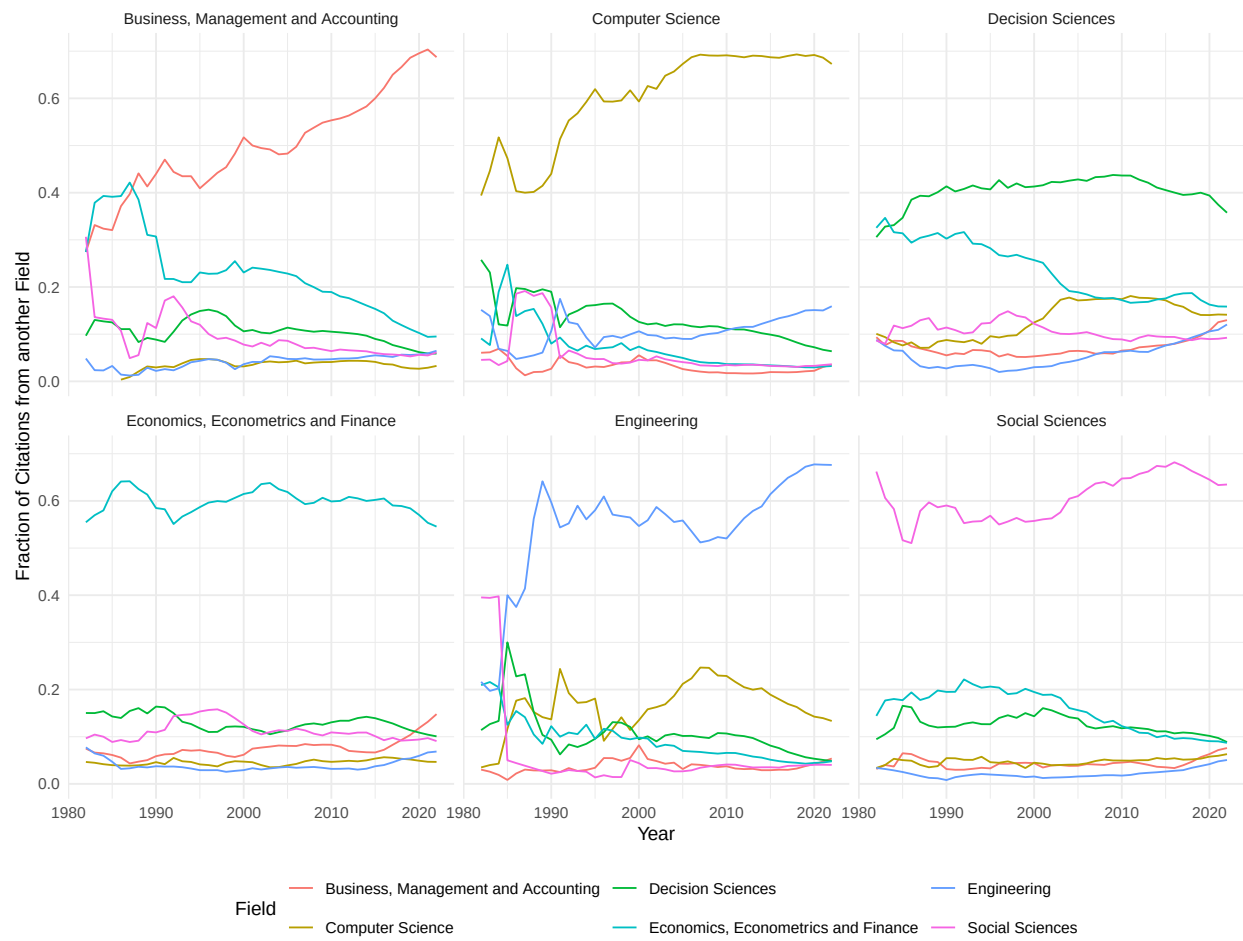
4.2.3 Per Window (Overlapping)



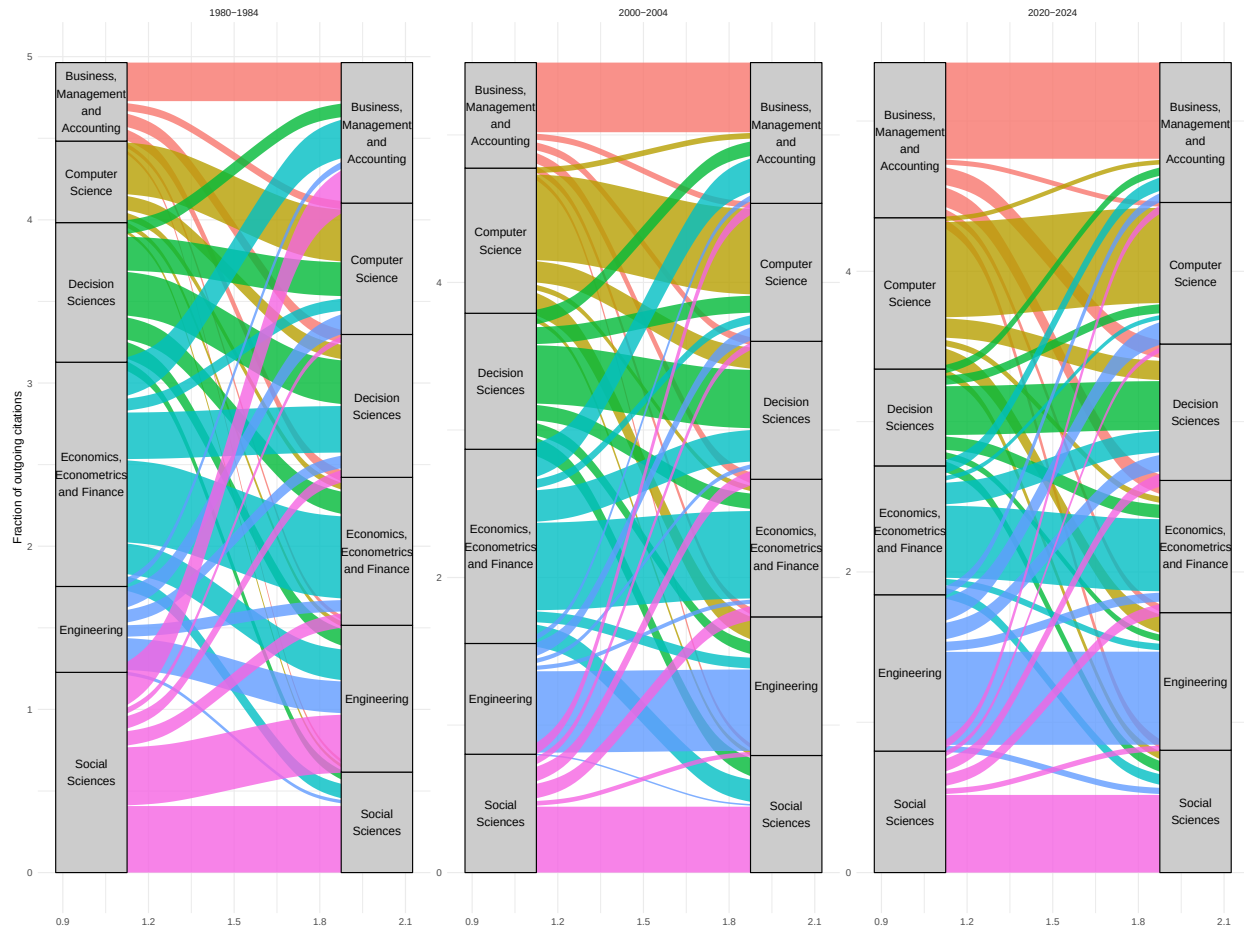
4.3 Citation Flows over Time



Fraction of Citations from a Field per 5-year Rolling Windows



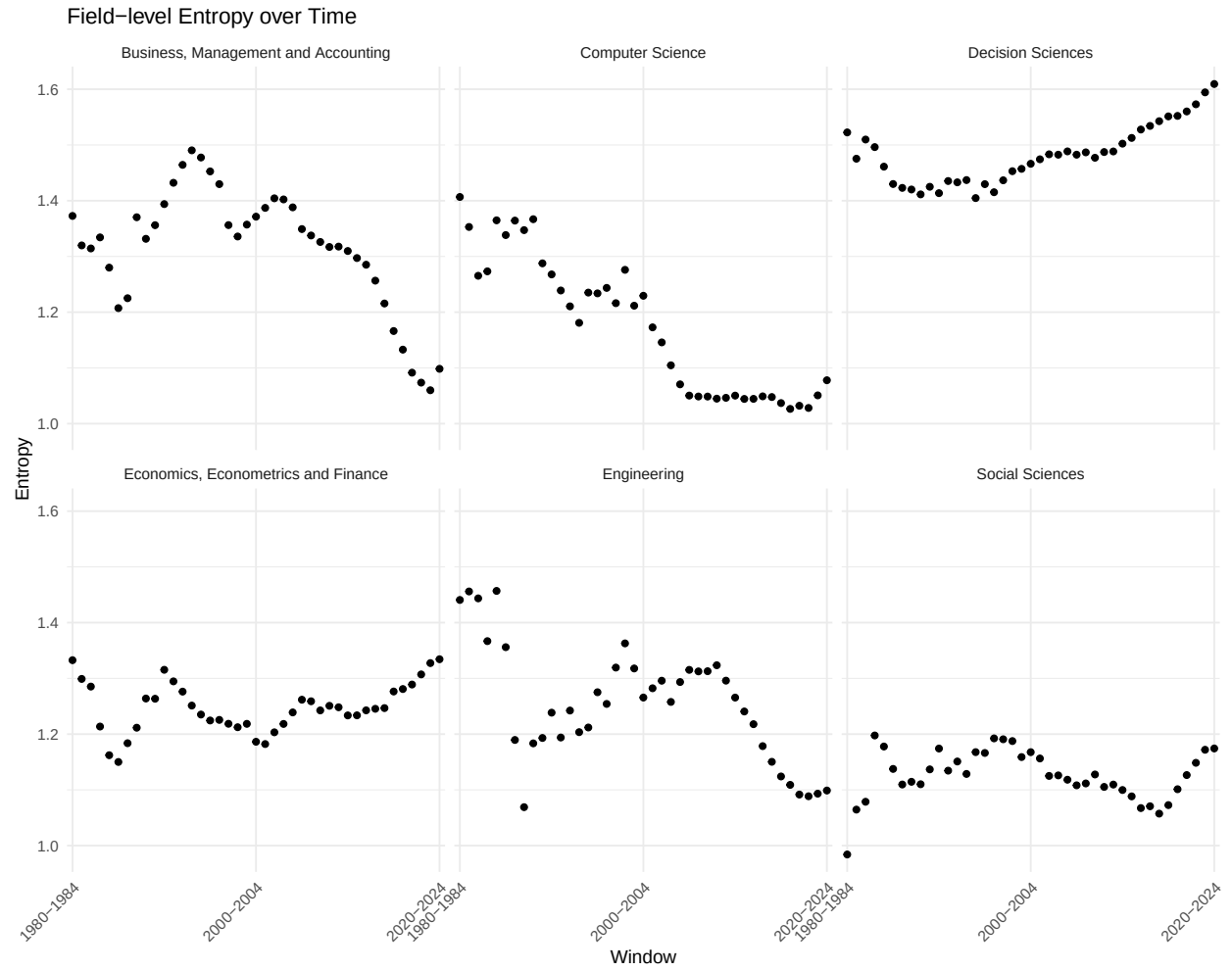
4.4 Alluvial Graph of Flows



4.5 Citation Flow Dispersion

Table 2: Citation Flow Entropy across Fields

Field	Flow Entropy
Business, Management and Accounting	53.88838
Computer Science	48.13214
Decision Sciences	60.76650
Economics, Econometrics and Finance	51.14528
Engineering	51.38705
Social Sciences	46.19559



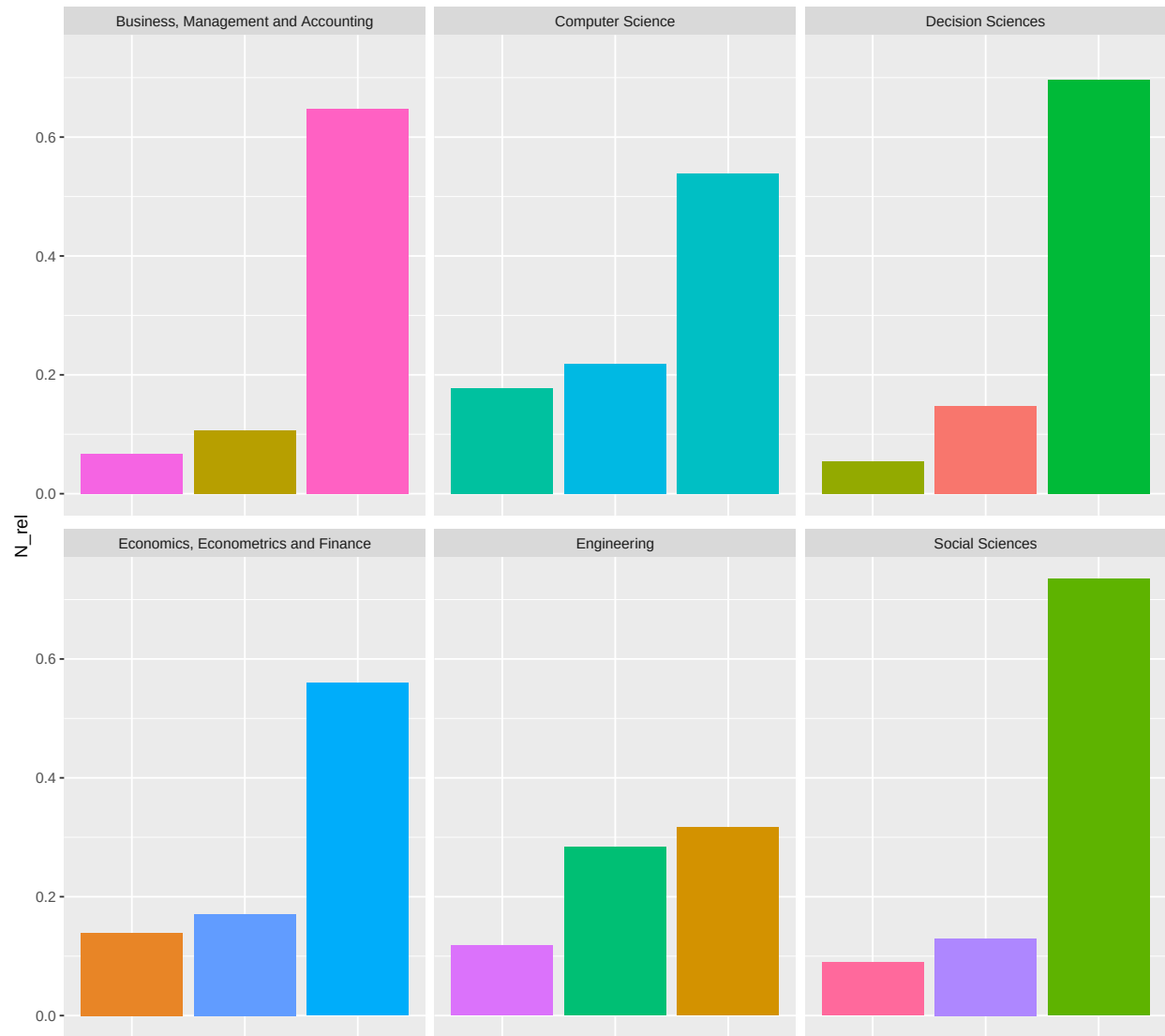
5. Topical Analysis of Fields

Table 3: Overview of Topics within Fields.

Field	Topic No.	Keywords	No. Articles
Business, Management and Accounting	0	supply, chain, game, manufacturer, retailer, channel, product, model, price, strategy	959
Business, Management and Accounting	1	evolutionary, government, green, enterprises, innovation, game, public, supervision, ppp, construction	158
Business, Management and Accounting	2	service, customers, game, queueing, equilibrium, games, server, allocation, queue, medical	100
Business, Management and Accounting	3	strategic, alliances, theory, alliance, knowledge, research, game, strategy, innovation, firms	92
Business, Management and Accounting	4	crowdfunding, financial, financing, lending, game, evolutionary, credit, p2p, venture, platform	55
Business, Management and Accounting	5	private, risk, project, ppp, projects, public, game, model, study, allocation	53
Business, Management and Accounting	6	audit, firms, reporting, auditing, tax, counterfeiting, enforcement, policy, environmental, game	42
Business, Management and Accounting	7	design, decentralized, problems, optimization, systems, designers, distributed, processes, solution, engineering	18
Business, Management and Accounting	8	mnes, location, internalization, choices, regionally, multinational, sites, competition, marketing, competing	3
Computer Science	0	game, users, network, networks, based, proposed, equilibrium, power, algorithm, spectrum	1686
Computer Science	1	games, game, agent, learning, agents, equilibrium, algorithm, based, nash, theory	684
Computer Science	2	game, security, defense, model, attack, data, attacks, network, privacy, attacker	554
Computer Science	3	blockchain, mining, miners, consensus, chain, supply, technology, game, transaction, pools	113
Computer Science	4	game, semantic, based, learning, image, word, rough, classification, feature, players	50
Computer Science	5	data, center, stable, reconfiguration, centers, cdg, virtual, electricity, resource, time	21
Computer Science	6	image, embedding, watermarking, denoising, pixels, steganography, cover, game, adaptive, jpeg	16
Computer Science	7	tracking, target, targets, mtt, motion, association, particle, visual, develop, estimation	4
Computer Science	8	bitc, bsns, body, operational, evolutionarily, cloud, data, conditions, configuration, intervals	4
Decision Sciences	0	game, games, equilibrium, nash, players, theory, equilibria, player, strategies, information	1052
Decision Sciences	1	auctions, market, auction, price, game, model, strategies, information, product, bidding	223
Decision Sciences	2	fuzzy, decision, game, cooperative, dea, interval, making, method, proposed, value	83
Decision Sciences	3	construction, project, projects, game, model, bidding, risk, theory, decision, contractors	83

Decision Sciences	4	innovation, government, evolutionary, technology, enterprises, carbon, digital, patent, strategy, model	49
Decision Sciences	5	brain, voting, turnout, models, decision, committees, committee, economics, choice, want	6
Decision Sciences	6	incentive, leader, follower, knowledge, mechanism, subordinates, transfer, employees, company, creation	5
Decision Sciences	7	dividend, risk, time, control, reinsurance, model, asset, problem, ruin, function	5
Decision Sciences	8	alert, citation, alertsnew, manage, added, profile, authors, 1145, hua, siteget	4
Economics, Econometrics and Finance	0	games, game, equilibrium, players, stochastic, value, cooperative, theory, paper, solution	745
Economics, Econometrics and Finance	1	model, firms, game, market, investment, equilibrium, competition, firm, innovation, amp	227
Economics, Econometrics and Finance	2	carbon, environmental, government, pollution, enterprises, evolutionary, green, game, climate, emission	185
Economics, Econometrics and Finance	3	monetary, sanctions, bank, policy, game, trade, fiscal, model, financial, risk	116
Economics, Econometrics and Finance	4	team, football, sports, game, soccer, play, theory, momentum, teams, league	37
Economics, Econometrics and Finance	5	compliance, tax, audit, trust, securities, taxpayers, des, litigation, accounting, la	8
Economics, Econometrics and Finance	6	field, mean, equations, algorithm, kolmogorov, problem, solving, discrete, fokker, parabolic	6
Economics, Econometrics and Finance	7	cattle, fed, conduct, oligopsony, power, market, meat, pricing, markets, processors	4
Economics, Econometrics and Finance	8	economic, arc, growth, course, human, retreat, advance, pressure, development, environment	4
Engineering	0	energy, market, game, power, electricity, model, based, demand, proposed, charging	827
Engineering	1	game, power, network, networks, allocation, users, interference, proposed, problem, algorithm	740
Engineering	2	security, game, attack, attacks, attacker, defender, cyber, physical, jamming, defense	309
Engineering	3	vehicle, vehicles, game, lane, traffic, driver, control, parking, driving, model	275
Engineering	4	game, control, pursuit, games, target, evasion, strategy, differential, nash, based	249
Engineering	5	water, port, game, model, shipping, theory, management, manufacturing, cooperative, strategy	161
Engineering	6	emergency, evacuation, crowd, pedestrians, model, game, pedestrian, government, behavior, exit	33
Engineering	7	gate, sizing, crosstalk, noise, interconnect, delay, optimization, wire, conflicting, simultaneous	14
Engineering	8	segmentation, image, landmark, detection, anomaly, shape, hyperspectral, clustering, game, clouds	6
Social Sciences	0	game, evolutionary, cooperation, games, theory, strategy, strategies, model, social, evolution	1290
Social Sciences	1	model, game, terrorism, political, terrorist, bargaining, policy, conflict, theoretic, international	228

Social Sciences	2	traffic, game, users, model, privacy, road, equilibrium, transportation, network, based	157
Social Sciences	3	reviews, live, streaming, marketing, consumers, online, commerce, quality, brand, product	31
Social Sciences	4	household, households, efficiency, matching, women, wives, marriage, husbands, couples, children	18
Social Sciences	5	investment, risk, optimal, insurance, reinsurance, asset, problem, pension, variance, company	10
Social Sciences	6	rumor, diffusion, rumors, propagation, image, media, pixels, disclosure, incidents, social	9
Social Sciences	7	violence, micro, accessed, 2008, http, felson, com, www, narrative, games	8
Social Sciences	8	swarm, unmanned, uav, autonomous, collaboration, mechanism, communication, strategy, dominance, evolutionary	4



Keywords

- | | |
|---|--|
| <ul style="list-style-type: none"> auctions, market, auction, price, game, model, strategies, information, product, bidding carbon, environmental, government, pollution, enterprises, evolutionary, green, game, climate, emission energy, market, game, power, electricity, model, based, demand, proposed, charging evolutionary, government, green, enterprises, innovation, game, public, supervision, ppp, construction fuzzy, decision, game, cooperative, dea, interval, making, method, proposed, value game, evolutionary, cooperation, games, theory, strategy, strategies, model, social, evolution game, games, equilibrium, nash, players, theory, equilibria, player, strategies, information game, power, network, networks, allocation, users, interference, proposed, problem, algorithm game, security, defense, model, attack, data, attacks, network, privacy, attacker | <ul style="list-style-type: none"> game, users, network, networks, based, proposed, equilibrium, power, algorithm, spectrum games, game, agent, learning, agents, equilibrium, algorithm, based, nash, theory games, game, equilibrium, players, stochastic, value, cooperative, theory, paper, solution model, firms, game, market, investment, equilibrium, competition, firm, innovation, amp model, game, terrorism, political, terrorist, bargaining, policy, conflict, theoretic, international security, game, attack, attacks, attacker, defender, cyber, physical, jamming, defense service, customers, game, queueing, equilibrium, games, server, allocation, queue, medical supply, chain, game, manufacturer, retailer, channel, product, model, price, strategy traffic, game, users, model, privacy, road, equilibrium, transportation, network, based |
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